

Cilium on Windows — eBPF Map Inspection Guide

Inspecting the eBPF-for-Windows (efw) datapath maps programmed by cilium-agent, using **bpftool.exe** and **ebpf_state.exe**. Generated for the AKS Windows node port.

Key concepts & gotchas

- **Maps are shared by pin path.** The agent and the datapath both pin maps under `/ebpf/global/<name>`. If the agent writes a map at a different path or under a different name than the loaded datapath programs read, the two never share the object.
- **Name truncation.** efw truncates the kernel object name to 15 characters (`BPF_OBJ_NAME_LEN=16`). So `bpftool map show` lists names like `cilium_lb4_serv.map` `dump name <fullname>` only works for names ≤ 15 chars; otherwise `dump` by numeric id.
- **Two tools.** `ebpf_state.exe` gives high-level decoded views (services, endpoints); `bpftool.exe` shows raw key/value bytes and works for every map (including CT/NAT that `ebpf_state` does not decode).
- **KubeProxyReplacement is off on this node:** ClusterIP, ExternalIPs and (via the Windows port) LoadBalancer VIPs are programmed into the datapath; NodePort frontends are not reflected while KPR is off.

1. Discover maps & resolve ids

```
# List every pinned map (id, type, name, key/value size, max entries)
bpftool.exe map show

# Filter to the maps you care about
bpftool.exe map show | Select-String
"lb4|ct4|ct_any4|lxc|cilium_ep|ipcache|reverse_nat|backend|affinity|snat|neigh"

# Resolve a (possibly truncated) name to its id, then dump it
$id = (bpftool.exe map show | Select-String "cilium_ct4_global").ToString().Split(":")[0].Trim()
bpftool.exe map dump id $id
```

2. Load-balancer maps (Service → Backend translation)

Populated by cilium-agent from Kubernetes Services/EndpointSlices. The XDP/TC datapath programs (`tail_lb_ipv4`, `cil_sock4_connect`, ...) read these to DNAT a Service VIP to a backend pod.

Map	What it stores	Why / When it is used
cilium_lb4_services	Frontend key (VIP+port+proto+scope) → backend slot / count / rev_nat index / service type.	The core Service lookup. Read on every connection to a ClusterIP / LoadBalancer / ExternalIP to pick a backend. Slot 0 = master entry (holds backend count & flags); slots 1..N = individual backends.
cilium_lb4_backends	Backend id → backend IP + port + proto + flags.	Resolves the backend slot chosen in the services map to the actual pod endpoint to DNAT to.
cilium_lb4_reverse_nat	rev_nat index → original VIP + port.	Un-DNAT: on the reply path, restores the client-visible Service VIP/port so responses look like they came from the VIP, not the pod.
cilium_lb4_affinity	(client, service) → last-picked backend + timestamp.	Only when sessionAffinity: ClientIP is set. Pins a client to the same backend within the affinity timeout.
cilium_lb4_reverse_nat_sock	socket-LB reverse-NAT state (connect-time LB).	Used by the socket-level LB (<code>cil_sock4_connect</code>) for connect()-time translation of ClusterIP for local pods.

```
# IPv4 (these names are <= 15 chars, so dump by name works)
bpftool.exe map dump name cilium_lb4_services
bpftool.exe map dump name cilium_lb4_backends
bpftool.exe map dump name cilium_lb4_reverse_nat
```

```
bpftool.exe map dump name cilium_lb4_affinity # only if sessionAffinity used
```

```
# High-level decoded view (frontend + backends together) via efw helper
```

```
ebpf_state.exe show loadbalancers filter 10.0.130.240 # ClusterIP
```

```
ebpf_state.exe show loadbalancers filter 20.69.171.89 # LoadBalancer VIP
```

```
# IPv6 equivalents
```

```
bpftool.exe map dump name cilium_lb6_services
```

```
bpftool.exe map dump name cilium_lb6_backends
```

```
bpftool.exe map dump name cilium_lb6_reverse_nat
```

3. Connection tracking (CT) maps

Track the state of every flow (TCP/UDP) so replies are allowed, NAT is consistent, and policy is only evaluated once per connection. Names exceed 15 chars — dump by id.

Map	What it stores	Why / When it is used
cilium_ct4_global	5-tuple → connection state (flags, backend id, timestamps, seen ports).	The main TCP/UDP conntrack table for service & policy connections. Read/updated on every packet to decide new-vs-established and which backend a flow is pinned to.
cilium_ct_any4_global	Non-TCP/UDP (ICMP, other) flow state.	Same purpose as ct4_global but for connectionless / 'any' protocols.
cilium_ct6_global / cilium_ct_any6_global	IPv6 equivalents.	IPv6 conntrack for TCP/UDP and other protocols respectively.

```
# Resolve ids first
```

```
bpftool.exe map show | Select-String "ct4_global|ct_any4|ct6_global|ct_any6"
```

```
# Dump by id (replace <id> from the line above)
```

```
bpftool.exe map dump id <id-of-cilium_ct4_global>
```

```
bpftool.exe map dump id <id-of-cilium_ct_any4_global>
```

```
# One-liner to dump the main v4 CT table
```

```
$ct = (bpftool.exe map show | Select-String "cilium_ct4_global").ToString().Split(":")[0].Trim()
```

```
bpftool.exe map dump id $ct | Select-String 'Found' # quick entry count
```

4. Endpoint / identity maps

Map	What it stores	Why / When it is used
cilium_lxc	Local endpoint (pod) key (IP/MAC) → endpoint id, security identity, MAC, ifindex.	The datapath's view of pods on THIS node. Used to attach the right policy/identity to traffic entering or leaving a local pod.
cilium_ep	Per-endpoint policy/metadata (cilium_ep_config style).	Endpoint configuration/state consumed by the per-endpoint BPF programs.
cilium_ipcache	IP/CIDR → security identity + tunnel endpoint.	Global IP→identity map for ALL cluster IPs (pods, nodes, world). Read to derive the identity of a remote peer for policy enforcement and routing/encapsulation.

```
bpftool.exe map dump name cilium_lxc # local pods / endpoints
```

```
bpftool.exe map dump name cilium_ep # endpoint metadata
```

```
bpftool.exe map dump name cilium_ipcache # IP -> identity (14 chars, name works)
```

```
# efw decoded endpoint view
```

```
ebpf_state.exe show endpoints filter 10.244.1.87
```

5. NAT, neighbor & misc datapath maps

Map	What it stores	Why / When it is used
cilium_snat_v4_external	SNAT state: (tuple) → translated source ip/port.	Source-NAT table for masquerading pod traffic leaving the node (and for NodePort/LB egress). Keeps SNAT mappings so replies are reverse-translated.
cilium_nodeport_neigh4	next-hop IP → MAC.	Neighbor cache the datapath uses to L2-forward NodePort/LB traffic to the correct next hop.
cilium_node_map / _v2	node id → node info.	Maps remote node identifiers used for routing and encryption decisions.
cilium_metrics	per-CPU datapath counters.	Drop/forward counters exported as Prometheus metrics; read by the agent, not on the hot path.
cilium_events / cilium_signal	perf event ring buffers.	Datapath → agent notifications (drops, trace, policy verdicts, signals). Consumed by monitor/Hubble.

```
bpftool.exe map show | Select-String "snat_v4|nodeport_neigh4|node_map|metrics"
bpftool.exe map dump id <id-of-cilium_snat_v4_external>
bpftool.exe map dump id <id-of-cilium_nodeport_neigh4>
```

6. Correlate maps with datapath programs

To confirm the datapath actually reads the maps the agent writes, list programs and the map ids they bind. If a program binds map id X, the agent must be writing map id X (not a differently-named duplicate).

```
bpftool.exe prog show
# e.g. tail_lb_ipv4 / cil_sock4_connect / cil_to_netdev list their map_ids;
# those ids must match cilium_lb4_services / cilium_lb4_backends from 'map show'.
```

7. Handy output options

```
bpftool.exe -j map dump id <id> # JSON
bpftool.exe -p map dump id <id> # pretty JSON
bpftool.exe map dump id <id> | Select-String 'Found' # entry count
bpftool.exe map lookup id <id> key <hex-bytes> # single key
bpftool.exe map dump name cilium_lb4_services | Measure-Object -Line
```

Reference: maps are pinned at /ebpf/global/<name>. v1 names (cilium_lb4_services / cilium_lb4_backends) are what the currently-loaded datapath binds; the agent programs them via a legacy-name alias so control plane and datapath share the same objects.